

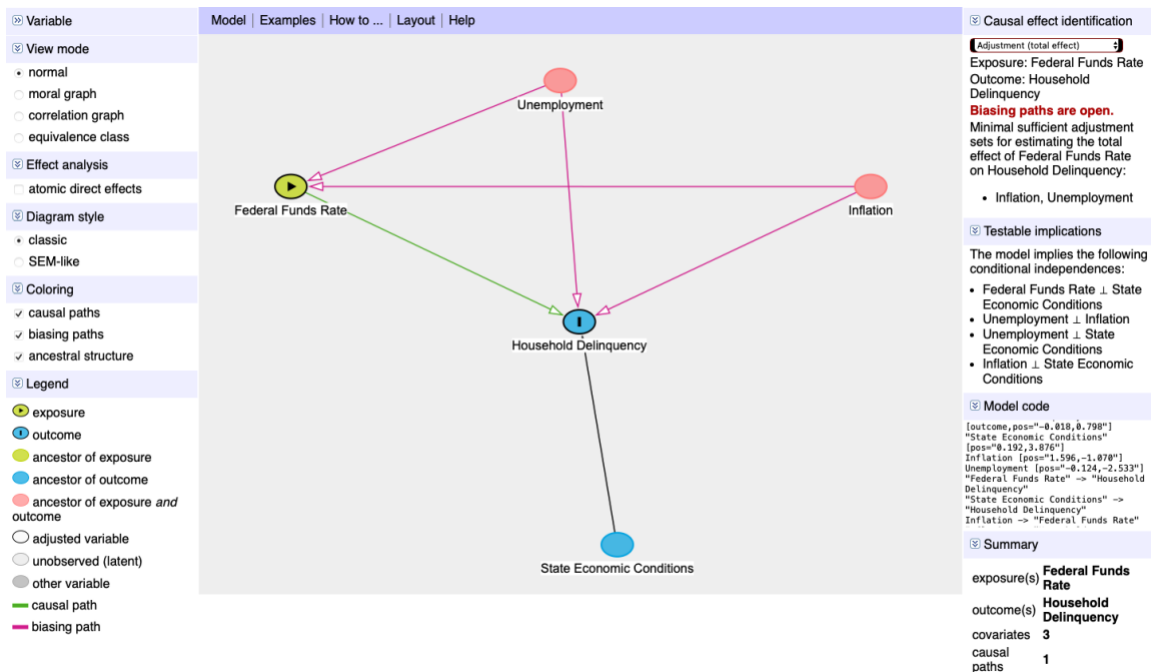
Empirical Strategy

In this project I am trying to look at how changes in interest rates might affect household credit delinquency. The main variable I focus on is the federal funds rate, since it reflects U.S. monetary policy. The outcome is household delinquency rates including the most widespread ones like credit cards, auto loans and mortgages measured at the state level.

The idea is that when interest rates increase, borrowing becomes more expensive and some households may have more difficulty making their payments on time. I use state-year panel data so I can look at how delinquency changes over time while accounting for differences across states.

DAG

I constructed a simple causal diagram (DAG) to visualize the relationships between the variables. In particular, macroeconomic factors like unemployment or inflation could affect both interest rates and delinquency, so they are possible confounders. State specific economic conditions may also influence delinquency outcomes. Here is my dag visualization:



Model

The baseline model is a panel regression with state and time fixed effects:

$$\text{Delinquency}_{st} = \beta_0 + \beta_1 \text{InterestRate}_t + \gamma_s + \delta_t + \varepsilon_{st}$$

Where:

- Delinquency_{st} = delinquency rate in state s at time t
- InterestRate_t = federal funds rate
- γ_s = state fixed effects
- δ_t = year fixed effects
- ε_{st} = error term

Delinquency is the state-level delinquency rate, and the federal funds rate captures changes in interest rates over time.

And also additional macroeconomic controls, such as unemployment or inflation, may also be included in the model. I am still undecided at this point though. Confounding factors has to be considered.

Assumptions / Possible Issues

One concern is that broader economic conditions like recessions may influence both monetary policy and delinquency at the same time. That could bias the estimates if not accounted for. To address this, the model includes time fixed effects and can also include controls such as unemployment or inflation. This doesn't solve everything of course, but it should help reduce some of the confounding.